

SATENAS, Sweden

WMO: 025200

Lat: 58.436N

Lon: 12.707E

Elev: 54

StdP: 100.68

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-14.0	-11.0	-15.8	0.9	-13.3	-13.1	1.2	-10.0	13.4	5.1	12.3	4.2	1.7	60	0.652

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.7	25.7	18.5	24.0	17.9	22.2	17.0	19.7	23.7	18.7	22.4	17.8	21.1	4.1	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.3	13.3	21.5	17.3	12.5	20.4	16.4	11.7	19.4	56.6	23.8	53.4	22.5	50.3	21.2	23.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.9	10.4	9.3	DB	-17.7	28.6	4.1	1.7	-20.6	29.9	-23.0	30.9	-25.3	31.8	-28.3	33.1	(4)
(5)			WB	-18.0	21.5	4.1	1.1	-20.9	22.3	-23.3	22.9	-25.5	23.5	-28.5	24.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.2	-1.3	-1.3	1.2	6.0	10.6	14.2	16.7	16.1	12.3	7.6	3.3	0.3
	DBStd	7.35	4.55	4.68	3.93	3.24	3.26	2.61	2.48	2.48	2.74	3.36	3.60	4.67
	HDD10.0	1697	349	315	272	127	32	1	0	0	10	89	200	302
	HDD18.3	4094	608	548	530	370	242	128	66	78	181	334	449	561
	CDD10.0	668	0	0	0	6	49	126	206	188	79	13	1	0
	CDD18.3	25	0	0	0	0	1	3	13	8	1	0	0	0
	CDH23.3	205	0	0	0	0	7	34	112	50	2	0	0	0
	CDH26.7	18	0	0	0	0	0	4	12	2	0	0	0	0
Wind	WSAvg	4.3	4.7	4.6	4.5	4.3	4.2	4.1	3.9	3.9	4.3	4.4	4.6	4.7
Precipitation	PrecAvg	713	55	44	35	44	48	77	73	72	68	73	62	62
	PrecMax	971	114	118	77	95	91	155	156	151	182	154	123	128
	PrecMin	413	11	6	9	5	6	16	3	13	8	11	21	17
	PrecStd	122	26	25	17	27	22	34	38	30	41	34	27	29
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.8	8.0	12.0	19.4	24.4	26.8	28.1	26.6	22.9	16.7	12.1	9.6
		MCWB	6.9	6.7	7.7	12.3	16.6	18.1	19.8	18.8	17.1	14.1	11.2	9.0
	2%	DB	6.5	6.6	9.6	16.1	21.3	23.8	26.2	24.8	20.2	14.5	10.7	8.3
		MCWB	5.7	5.6	6.4	10.7	14.7	17.1	19.0	18.5	15.8	12.9	9.8	7.6
	5%	DB	5.4	5.6	7.8	13.6	18.8	21.7	24.5	23.0	18.3	13.4	9.7	7.4
		MCWB	4.6	4.8	5.4	9.2	13.6	15.9	18.4	17.7	15.1	12.1	8.9	6.7
	10%	DB	4.4	4.7	6.3	11.5	16.5	19.7	22.5	21.1	16.8	12.4	8.5	6.2
		MCWB	3.7	3.9	4.6	8.1	12.2	14.9	17.6	17.1	14.4	11.2	7.7	5.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.3	7.1	8.5	13.4	17.8	19.6	21.5	20.7	18.2	14.6	11.5	9.2
		MCDB	7.8	7.7	10.7	17.7	22.4	24.1	25.7	24.2	20.8	16.2	12.0	9.4
	2%	WB	5.9	5.9	7.0	11.2	15.7	17.9	20.2	19.3	16.6	13.3	10.0	7.9
		MCDB	6.4	6.3	8.9	15.0	19.8	22.6	24.4	22.7	19.1	14.1	10.4	8.3
	5%	WB	4.8	5.0	5.9	9.8	14.2	16.7	19.2	18.5	15.5	12.4	8.9	6.8
		MCDB	5.2	5.4	7.3	12.9	17.5	20.4	23.1	21.7	17.7	13.2	9.5	7.2
	10%	WB	3.8	4.0	4.9	8.6	12.9	15.6	18.1	17.6	14.6	11.4	7.9	5.6
		MCDB	4.3	4.7	6.1	11.0	15.9	19.0	21.6	20.4	16.5	12.3	8.5	6.1
Mean Daily Temperature Range	5% DB	MDBR	4.4	4.8	6.4	8.1	8.6	8.5	8.7	8.1	7.2	5.8	4.2	4.4
		MCDBR	4.3	4.6	8.3	12.1	12.6	12.4	12.6	11.2	9.5	6.3	4.7	4.3
		MCWBR	4.2	4.0	5.8	7.4	7.1	6.8	6.6	6.1	6.0	4.9	4.6	4.3
	5% WB	MCDBR	4.1	4.1	7.5	10.7	10.8	11.0	10.8	9.4	8.5	5.9	4.5	4.4
		MCWBR	4.1	3.8	5.7	6.9	6.7	6.6	6.3	5.8	5.8	5.0	4.5	4.5
Clear-Sky Solar Irradiance	taub	0.279	0.314	0.334	0.373	0.369	0.358	0.379	0.375	0.345	0.333	0.301	0.263	
	taud	2.236	2.311	2.374	2.325	2.397	2.448	2.409	2.419	2.492	2.441	2.310	2.170	
	Ebn at Noon	556	715	812	833	864	879	848	824	798	690	526	441	
	Edh at Noon	46	70	86	107	106	103	105	97	77	62	43	35	
All-Sky Solar Radiation	RadAvg	0.34	0.91	2.43	3.91	5.16	5.80	5.18	4.09	2.78	1.25	0.45	0.24	
	RadStd	0.06	0.15	0.23	0.49	0.44	0.31	0.66	0.37	0.33	0.20	0.08	0.02	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.46	N/A	N/A	N/A	N/A	N/A	-126	-177	N/A	N/A
(47) Regional (55 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page